

Areas of Competencies for National Skill Competency Test of IT Graduates

Areas of Competencies	Weightage %	Topics
Computer Networks and Cloud Computing	10%	1 - Data Communication
		2 - Computer Networks
		3 - Data Link Layer
		4 - Network Layer
		5 Transport Layer
		6 - Application Layer
		7 - Wireless Networks
		8 - Cloud Computing
		9 - Network Security (Networks Perspective)
		10 - Next Generation Networks
Programming (C++/Java/Python)	10%	1. Programming Fundamentals
		2. Data Types & Variables
		3. Operators & Expressions
		4. Control Structures
		5. Functions / Methods,
		6. Input / Output Handling.
		7. Strings & Text Processing
		8. Arrays & Collections
		9. Object-Oriented Programming (OOP)
		10. Memory Management Concepts
		11. Exception & Error Handling
		12. Modules, Packages & Libraries
		13. Advanced Programming Concepts
		14. Concurrency & Parallelism (Introductory)
		15 Debugging, Testing & Optimization
		16. Software Development Practices
Data Structures & Algorithms	10%	1 - Foundations of Data Structure and Algorithms
		2 - Linear Data Structures
		3 - Non-Linear Data Structures
		4 - Searching Algorithms
		5 - Sorting Algorithms
		6 - Hashing
		7 - Tree Algorithms
		8 - Graph Algorithms
		9 - Algorithm Design Techniques
		10 - Advanced Data Structures
		11 - String Algorithms
		12 - Complexity & Optimization
Operating Systems	5%	1. Introduction to Operating Systems
		2. Operating System Structures
		3. Process Management
		4. CPU Scheduling
		5. Thread Management
		6. Concurrency & Synchronization
		7. Deadlocks
		8. Memory Management
		9. File System Management
		10. Secondary Storage Management
		11. Input / Output Systems
		12. Protection & Security

Areas of Competencies	Weightage %	Topics
Software Engineering	10%	1. Introduction to Software Engineering
		2 Software Process Models
		3. Agile Software Development
		4. Software Requirements Engineering
		5. Software Project Management
		6. Software Design
		7. Software Architecture
		8. User Interface Design
		9. Software Implementation & Coding
		10. Software Testing
		11. Software Maintenance & Evolution
		12. Software Quality Assurance
		13. Software Metrics & Measurement
		14. Software Configuration Management
		15. Software Risk Management
		16. Software Security Engineering
Web Development	10%	1. Introduction to Web Development
		2. Web Architecture & Protocols
		3. HTML Fundamentals
		4. CSS Fundamentals
		5. Advanced CSS & Responsive Design
		6. JavaScript Fundamentals
		7. Advanced JavaScript
		8. Frontend Frameworks & Libraries
		9. Backend Development Fundamentals
		10. Server-Side Programming
		11. Databases for Web Applications
		12. Web Security
		13. Web Performance & Optimization
		14. Web Testing & Debugging
		15. Deployment & Hosting
		16. Web APIs & Integration
		17. Modern Web Development Practices
AI / Machine Learning and Data Analytics	10%	1. Introduction to AI, ML & Data Analytics
		2. Mathematical Foundations
		3. Python for AI & Data Analytics
		4. Data Collection & Pre-processing
		5. Exploratory Data Analysis (EDA)
		6. Supervised Learning
		7. Ensemble Learning
		8. Unsupervised Learning
		9. Model Evaluation & Validation
		10. Feature Engineering & Selection
		11. Deep Learning Fundamentals
		12. Advanced Deep Learning
		13. Natural Language Processing (NLP)
		14. Computer Vision
		15. Big Data Analytics (Introductory)
		16. Model Deployment & MLOps Basics
		17. AI Ethics, Security & Privacy
Cyber Security	5%	1. Introduction to Cyber Security
		2. Security Fundamentals & Principles
		3. Cryptography Basics
		4. Network Security

Areas of Competencies	Weightage %	Topics
		5. Operating System Security
		6. Web Application Security
		7. Malware & Attack Techniques
		8. Authentication & Access Control
		9. Secure Software Development
		10. Wireless & Mobile Security
		11. Cloud & Virtualization Security
		12. Digital Forensics
		13. Incident Response & Management
		14. Security Monitoring & Auditing
		15. Cyber Laws & Ethics
		16. Emerging Trends in Cyber Security
Databases	10%	1. Introduction to Database Systems
		2. Database System Architecture
		3. Data Models
		4. Relational Database Concepts
		5. Relational Algebra & Calculus
		6. Structured Query Language (SQL)
		7. Advanced SQL
		8. Database Design & Normalization
		9. Transaction Management
		10. Concurrency Control
		11. Recovery Management
		12. Indexing & File Organization
		13. Query Processing & Optimization
		14. Database Security
		15. Distributed Databases
		16. NoSQL & Modern Databases
		17. Data Warehousing & Data Mining (Introductory)
Problem Solving And Analytical Skills	20%	1. Introduction to Problem Solving
		2. Problem Understanding & Analysis
		3. Logical Reasoning Fundamentals
		4. Algorithms & Flow Control
		5. Data Representation & Abstraction
		6. Pattern Recognition & Generalization
		7. Mathematical & Quantitative Reasoning
		8. Algorithmic Thinking
		9. Critical Thinking & Decision Making
		10. Debugging & Error Analysis
		11. Complexity & Efficiency Awareness
		12. Problem Solving Using Programming
		13. Data-Driven Problem Solving
		14. Creative & Innovative Thinking
		15. Real-World Problem Solving
		16. Communication & Documentation of Solutions